

Market Brain Grid (MBG) — Product Requirements Document

Autonomous Multi-Agent Quant Trading, Econometric Proofing & Verifiable Execution Architecture

Codename / Title: Market Brain Grid (MBG)	Agent Engine: Hermes Agent CLI / Docker Swarm Runner	Backend / Edge: Supabase (Postgres + pgvector) + Edge Webhooks	Target Universe: IDX Equities (LQ45/IDX80) + Crypto Perps (Hyperliquid/OKX)
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1. Executive Summary & Core Mission

Market Brain Grid (MBG) is an institutional-grade, multi-agent algorithmic intelligence and execution framework designed to operate seamlessly across the **Indonesia Stock Exchange (IDX)** and **Cryptocurrency Perpetuals markets**. Powered by the **Hermes Agent** engine, MBG eliminates discretionary cognitive fatigue by replacing manual charting and linear prompt checklists with an autonomous 7-bot topology operating on a verified closed-loop execution graph.

MBG incorporates the rigor of cutting-edge academic and production frameworks: the multi-agent adversarial debate protocol from *TradingAgents* (arXiv:2412.20138), the 5-pillar quant factor decomposition and weekly reflection loop from *Moss AI*, strict point-in-time and lookahead guards from *HKUDS Vibe-Trading*, closed-loop stage evaluation from *CLQT* (arXiv:2606.29771), and econometric strategy validation from *Bailey & López de Prado* (DSR, CSCV, Hansen SPA).

2. The 5-Stage Closed-Loop Operating Pipeline (CLQT Standard)

MBG executes through 5 strictly enforced stages, eliminating data snooping and decision drift:

- Stage 1 — Intelligence Ingestion (II):** Point-in-Time feeds, multi-depth orderbooks, IDX broker summaries, and perpetual funding rates with strict NaN fail-closed validation.
- Stage 2 — Multi-Pillar Quantitative Synthesis (MPQS):** 5-Pillar signal decomposition (Trend, Momentum, Reversion, Volume, Volatility) combined with Smart Money Concepts (FVG, Order Blocks, Liquidity Sweeps).
- Stage 3 — Adversarial Duel & Econometric Proofing (ADEP):** Forced Bull vs. Bear debate with catastrophic failure analysis, followed by independent mathematical refereeing ($DSR \geq 0.95$, $PBO \leq 0.15$).
- Stage 4 — Risk Mandate & Position Sizing (RMPS):** Volatility-adjusted ATR and fractional Kelly sizing; automated portfolio daily drawdown ceiling locked at 2.5%.
- Stage 5 — Verifiable Execution & Hash Ledger (VEHL):** Emits deterministic execution tickets, binds decisions into an immutable SHA-256 hash chain, and triggers alerts/order routing.

3. Market Brain Grid (MBG) Swarm Persona Architecture (7-Bot Topology)

Bot / Persona	Operational Mandate	Deterministic Tools Executed	Output Schema & Invariant
1. MBG-DataScout	Point-in-Time data cleaner, tick streamer, trading halt filter, purged/embargoed splitting.	fetch_idx_clean(), fetch_crypto_ohlcv(), fetch_orderbook_depth().	NormalizedCandleBuffer (Strict NaN fail-closed; zero forward-looking leakage).
2. MBG-QuantAlpha	5-Pillar factor computation, Smart Money Concepts (FVG, OB), market regime tagging.	TA-Lib, Formulaic Alpha DSL, SMC detector (Order Blocks, Liquidity Sweeps).	AlphaSignalMatrix (Pillar scores normalized -100 to +100).
3. MBG-IDXSpecialist	IDX Bandarmology (Top 3 net accumulation), Foreign Institutional Flow, ARA/ARB bounds.	fetch_broker_summary(), check_uma_status(), get_idx_tick_size().	IDXMicrrostructureReport (Accumulation status, broker concentration %, tick size).
4. MBG-CryptoSpecialist	Perpetuals funding rate drag, open interest divergence, liquidation cluster heatmaps.	fetch_funding_rate(), fetch_open_interest(), fetch_liquidation_clusters().	CryptoDerivativesReport (Funding bias score, squeeze risk, liquidation targets).
5. MBG-EconometricProofer	Independent mathematical referee running DSR, CSCV, PBO,	scipy.stats, CSCV combinatorial matrix engine, Hansen bootstrap tester.	ProofingCertificate ($DSR \geq 0.95$, $PBO \leq 0.15$; holds hard veto power).

Bot / Persona	Operational Mandate	Deterministic Tools Executed	Output Schema & Invariant
	and Hansen SPA.		
6. MBG-DebateArbiter	Adversarial stress-testing. Hosts structured Bull vs. Bear duel with mandatory tail-risk cases.	Multi-turn LLM debate harness, Invalidation criteria matrix, Loss taxonomy DB.	DebateVerdict (Confidence tier: High/Med/Low, Consensus Thesis, Invalidation Price).
7. MBG-RiskGuard	ATR volatility sizing, Daily Drawdown Lock ($\leq 2.5\%$), SHA-256 Audit Hash Chain generation.	calculate_atr_sizing(), check_max_drawdown(), compute_audit_hash().	ExecutableTradingPlan (Entry, Hard SL, TP1, TP2, Max Lots/Contracts, SHA-256 Hash).

4. Market Microstructure Mechanics & Friction Modeling

4.1 Indonesia Stock Exchange (IDX) Equity Microstructure

- Tick Matrix Enforcement:** Enforces Rp 1, 2, 5, 10, 25 tiers. Backtests simulating fills at non-tick increments are disqualified.
- ARA/ARB Boundary Modeling:** Auto-rejection limits dynamically freeze order fills if price hits boundary. Forbids unrealistic backtest fills when an equity is locked at upper limit (ARA).
- Bandarmology & Institutional Tracking:** Analyzes Top 3 Buyer/Seller volume share (Broker Summary). If Top 3 buyer volume > 65% of total session volume while price consolidates within 1.5% ATR, trigger *Accumulation Accumulator Signal*.
- Settlement & Cost Friction:** Models IDX T+2 settlement for cash availability. Friction: 0.15% Buy Commission, 0.25% Sell Commission (includes IDX levy and sales tax).

4.2 Crypto Perpetuals Microstructure (Hyperliquid / Binance / OKX)

- Funding Rate Dynamic:** High positive funding ($> +0.03\%$ / 8hr) with rising Open Interest signals over-leveraged longs → Bull setups are de-prioritized; long squeeze traps are flagged.
- Execution Friction:** Models 0.02% maker, 0.05% taker fees, plus dynamic slippage based on top-5 book depth.
- Cross-Margin & Liquidation Shield:** Positions automatically calculate liquidation price distance; minimum liquidation buffer must exceed $3.5 \times$ Daily ATR.

5. Econometric Strategy Proofing & In-Context Evolution

- Deflated Sharpe Ratio (DSR ≥ 0.95):** Penalizes strategy return by trial count $\$N$ logged in the TrialLedger, adjusting for skewness and kurtosis.
- Combinatorially Symmetric Cross-Validation (PBO ≤ 0.15):** Slices history into 16 blocks ($\$ \text{inom}\{16\}\{8\} = 12,870\$$ combinatorial splits) to guarantee the strategy does not degrade out of sample.
- Contrastive Meta-Evolution (AlgoEvolve Standard):** Weekly tuning injects the Top-2 performing strategies and Top-2 worst failing strategies into context. Parameter adjustments are restricted to $\pm 30\%$ bounded drift around tactical thresholds, keeping core strategy logic immutable.

6. Standardized Daily Output Contract (Audit-Hash Linked)

Every morning at 08:15 WIB (IDX) and at every 8-hour funding cycle (Crypto), MBG delivers an actionable, verifiable trade plan:

```
{
  "plan_id": "MBG-IDX-BBCA-20260908",
  "generated_at": "2026-09-08T08:15:00+07:00",
  "audit_hash_chain": {
    "prior_round_action_hash": "a1b2c3d4e5f6...7890",
    "current_observation_hash": "e9f8d7c6b5a4...1234",
    "current_action_hash": "8f7a6b5c4d3e...5678"
  },
  "symbol": "BBCA.JK",
  "market_type": "IDX_EQUITY",
  "regime": "ACCUMULATION_BREAKOUT",
  "bias": "LONG",
  "confidence_level": "High Confidence",
  "quant_analysis": {
    "trend_pillar": "Bullish EMA cross (EMA20 > EMA50), Price above 200 EMA",
    "momentum_pillar": "RSI(14) 58.4 upward trajectory, MACD histogram expanding",
    "reversion_pillar": "Bollinger Bands contracting; volatility squeeze imminent",
    "volume_pillar": "Volume 2.1x 20-day MA on prior session bounce",
    "specialist_flow": "Foreign Net Buy +Rp 84B. Top brokers CC & AK absorbed 62% of supply"
  },
  "adversarial_scrutiny": {
    "bull_thesis": "Clear break of Rp 10,150 consolidation box backed by institutional foreign inflow",
    "bear_rebuttal": "Overhead supply zone at Rp 10,700 from July distribution block",
    "verdict_mitigant": "Risk/reward to first resistance is 1:2.45; tight stop below support mitigates risk"
  },
  "econometric_proofing": {
    "deflated_sharpe_ratio_dsr": 0.971,
    "probability_of_backtest_overfitting_pbo": 0.084,
    "walk_forward_efficiency_wfe": 0.72,
    "hansen_spa_pvalue": 0.024
  },
}
```

```

"trade_execution": {
  "entry_strategy": "LIMIT_ORDER_TRANCHE",
  "entry_range": [10150, 10250],
  "hard_stop_loss": 9850,
  "take_profit_target_1": 10700,
  "take_profit_target_2": 11150,
  "risk_reward_ratio": 2.45,
  "max_nav_allocation_pct": 5.0,
  "calculated_lots": 45
},
"invalidation_rules": [
  "Cancel pending limit orders if opening gap exceeds +2.5% (ARA chase forbidden)",
  "Hard cut if price closes below Rp 9,850 on 1-hour candle"
]
}

```

7. Infrastructure Blueprint & Implementation Roadmap

Hybrid Architecture (Local Compute + Supabase State)

- **Local Host (CLI/Docker):** Executes the Hermes CLI swarm runner, TA calculations, backtesting loops, and LLM reasoning. Zero recurring GPU/server overhead.
- **Supabase (PostgreSQL + pgvector):** Persists the TrialLedger, the audit hash chain, daily trade plans, and strategy memory vectors.

1. **Phase 1: Proofing & Math Engine:** Implement `dsr.py` (Bailey & López de Prado formula), `cscv.py`, and `audit_hash.py`.
2. **Phase 2: Point-in-Time Data Feeds:** Build CCXT perp streaming and IDX scrapers with strict purged/embargoed split logic and NaN fail-closed validation.
3. **Phase 3: Multi-Agent Debate & Arbiter Swarm:** Configure Hermes Agent with system prompts enforcing forced bear rebuttals, 6-call loop limits, and JSON output parsing.
4. **Phase 4: Cron Runner & Daily Proofed Delivery:** Schedule automated 08:15 WIB scans for IDX and 8-hour funding alerts for Crypto, pushing certified Markdown cards directly to Telegram.

Non-Negotiable Execution Guardrails

- **Circuit-Breaker Daily Drawdown:** If cumulative portfolio NAV falls > 2.5% in 24 hours, all open orders are canceled and trading halts for 24 hours.
- **Risk-per-Trade Cap:** Maximum dollar loss per trade must not exceed 1.25% of active equity.
- **Unanimous Gatekeeper Approval:** If MBG-EconometricProofer ($DSR < 0.95$) or MBG-RiskGuard flags a violation, the trade is rejected unconditionally.